

Experiment-5

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In [1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Step 1: Load Dataset
df = pd.read_csv(
    "Sample - Superstore.csv",
    encoding="latin1"
)
print("Dataset Loaded Successfully")
print(df.head())

# Step 2: Check Dataset
print("\nDataset Shape:")
print(df.shape)
print("\nColumn Names:")
print(df.columns.tolist())

# Step 3: Verify Data
print("\nData Types:")
print(df.dtypes)
print("\nMissing Values:")
print(df.isnull().sum())

# Step 4: Calculate KPIs
total_sales = df["Sales"].sum()
total_profit = df["Profit"].sum()
total_orders = df["Order ID"].nunique()
average_order_value = total_sales / total_orders
print("\n===== KPIs =====")
print("Total Sales:", round(total_sales, 2))
print("Total Profit:", round(total_profit, 2))
print("Total Orders:", total_orders)
print("Average Order Value:", round(average_order_value, 2))

# Step 5: Bar Chart - Sales by Region
region_sales = df.groupby("Region")["Sales"].sum()
plt.figure(figsize=(8, 5))
region_sales.plot(kind="bar")
plt.title("Sales by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()

# Step 6: Line Chart - Monthly Sales
df["Order Date"] = pd.to_datetime(
    df["Order Date"],
    format="mixed",
    errors="coerce"
)
monthly_sales = df.groupby(
    df["Order Date"].dt.to_period("M")
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)["Sales"].sum()
plt.figure(figsize=(10, 5))
monthly_sales.plot(
    kind="line",
    marker="o"
)
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# Step 7: Pie Chart - Sales by Category
category_sales = df.groupby("Category")["Sales"].sum()
plt.figure(figsize=(7, 7))
category_sales.plot(
    kind="pie",
    autopct="%1.1f%%"
)
plt.title("Sales by Category")
plt.ylabel("")
plt.tight_layout()
plt.show()

# Step 8: Correlation Heatmap
numeric_data = df.select_dtypes(
    include="number"
)
correlation = numeric_data.corr()
plt.figure(figsize=(9, 6))
sns.heatmap(
    correlation,
    annot=True,
    fmt=".2f",
    cmap="coolwarm"
)
plt.title("Correlation Heatmap")
plt.tight_layout()
plt.show()

# Step 9: Filter / Slicer
print("\nAvailable Regions:")
print(df["Region"].unique())
region = input(
    "\nEnter Region (East/West/North/South): "
)
filtered_data = df[
    df["Region"].str.lower() == region.lower()
]
print("\n===== Filtered Data =====")
print(filtered_data.head())
filtered_sales = filtered_data["Sales"].sum()
filtered_profit = filtered_data["Profit"].sum()
print("\nFiltered Sales:",
      round(filtered_sales, 2))
print("Filtered Profit:",
      round(filtered_profit, 2))

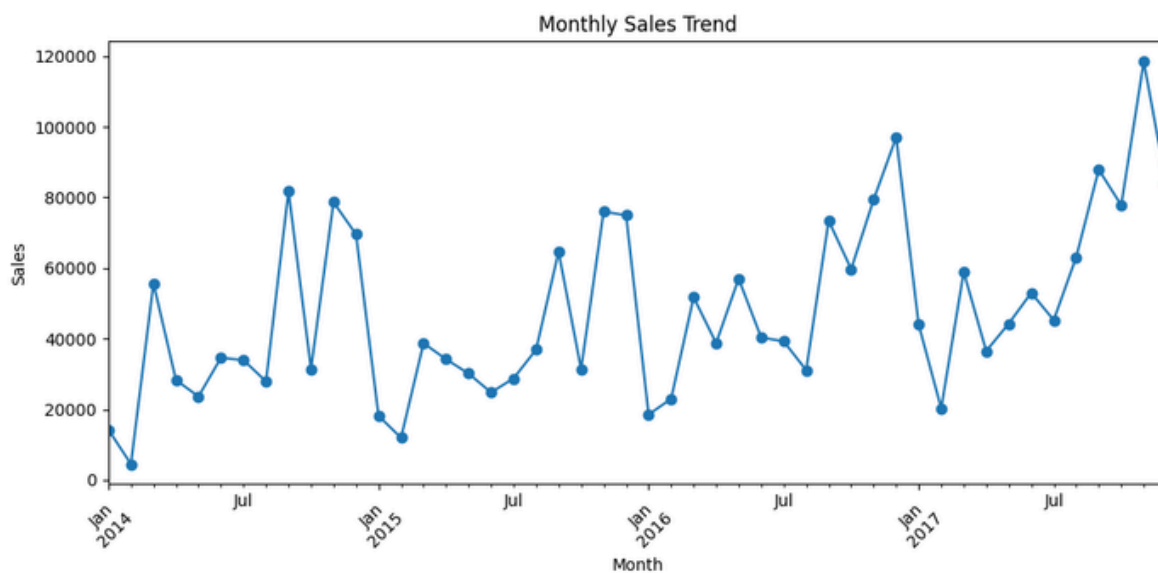
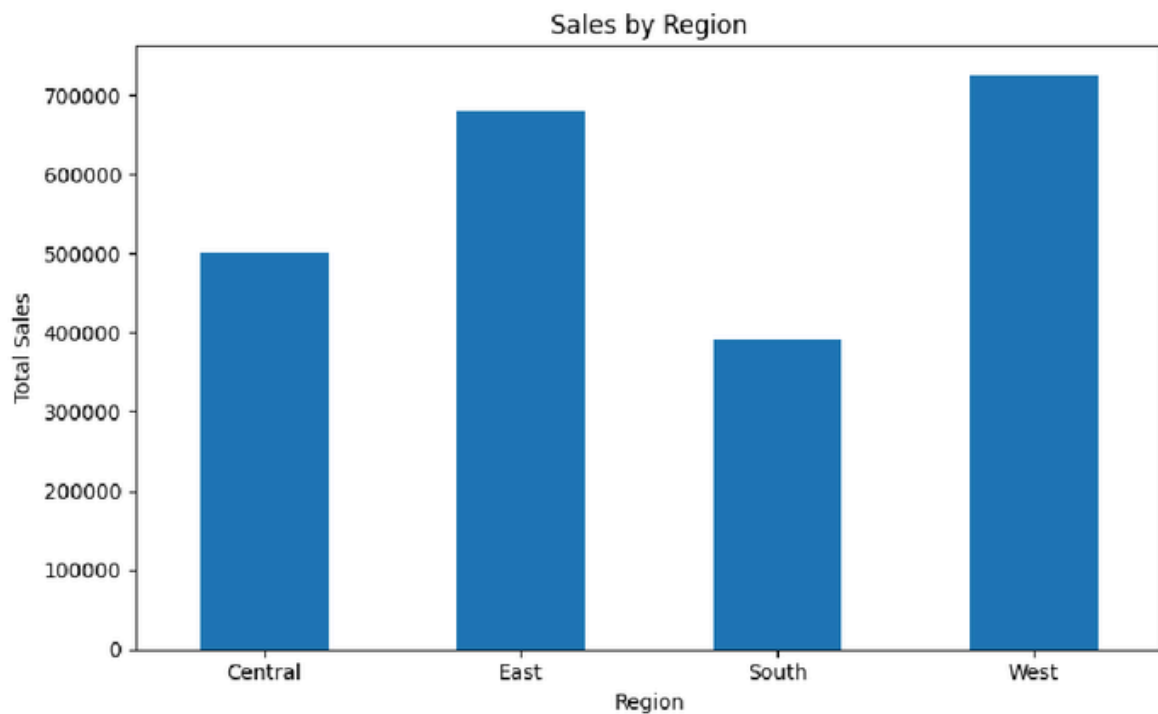
# Step 10: Business Insights

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print("\n===== Business Insights =====")
top_region = region_sales.idxmax()
top_category = category_sales.idxmax()
print("Top Sales Region:", top_region)
print(
    "Highest Region Sales:",
    round(region_sales.max(), 2)
)
print("Top Sales Category:", top_category)
print(
    "Highest Category Sales:",
    round(category_sales.max(), 2)
)

```



Dataset Loaded Successfully

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID \
0	1	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520
1	2	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520
2	3	CA-2016-138688	06-12-2016	6/16/2016	Second Class	DV-13045
3	4	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335
4	5	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335

	Customer Name	Segment	Country	City ... \
0	Claire Gute	Consumer	United States	Henderson ...
1	Claire Gute	Consumer	United States	Henderson ...
2	Darrin Van Huff	Corporate	United States	Los Angeles ...
3	Sean O'Donnell	Consumer	United States	Fort Lauderdale ...
4	Sean O'Donnell	Consumer	United States	Fort Lauderdale ...

	Postal Code	Region	Product ID	Category	Sub-Category \
0	42420	South	FUR-BO-10001798	Furniture	Bookcases
1	42420	South	FUR-CH-10000454	Furniture	Chairs
2	90036	West	OFF-LA-10000240	Office Supplies	Labels
3	33311	South	FUR-TA-10000577	Furniture	Tables
4	33311	South	OFF-ST-10000760	Office Supplies	Storage

	Product Name	Sales	Quantity \
0	Bush Somerset Collection Bookcase	261.9600	2
1	Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.9400	3
2	Self-Adhesive Address Labels for Typewriters b...	14.6200	2
3	Bretford CR4500 Series Slim Rectangular Table	957.5775	5
4	Eldon Fold 'N Roll Cart System	22.3680	2

	Discount	Profit
0	0.00	41.9136
1	0.00	219.5820
2	0.00	6.8714
3	0.45	-383.0310
4	0.20	2.5164

[5 rows x 21 columns]

Dataset Shape:
(9994, 21)

Column Names:

['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode', 'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'State', 'Postal Code', 'Region', 'Product ID', 'Category', 'Sub-Category', 'Product Name', 'Sales', 'Quantity', 'Discount', 'Profit']

Data Types:

Row ID	int64
Order ID	object
Order Date	object
Ship Date	object
Ship Mode	object
Customer ID	object
Customer Name	object
Segment	object
Country	object
City	object
State	object
Postal Code	object
	int64

Region		object
Product	ID	object
Category	Sub-	object
Category		object
Product	Name	object
Sales		float64
Quantity		int64
Discount		float64
Profit	dtype:	float64
		object

Missing Values:

Row ID		0
Order ID		0
Order Date	Ship	0
Date		0
Ship	Mode	0
Customer	ID	0
Customer	Name	0
Segment	Country	0
City		0
State		0
Postal	Code	0
Region		0
Product	ID	0
Category	Sub-	0
Category		0
Product	Name	0
Sales		0
Quantity		0
Discount	Profit	0
dtype: int64		0
		0

===== KPIs =====

Total	Sales:	2297200.86
Total	Profit:	286397.02
Total	Orders:	5009
Average	Order Value:	458.61

Available Regions:

['South' 'West' 'Central' 'East']

===== Filtered Data =====

Empty DataFrame

Columns: [Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Country, City, State, Postal Code, Region, Product ID, Category, Sub-Category, Product Name, Sales, Quantity, Discount, Profit]

Index: []

[0 rows x 21 columns]

Filtered Sales: 0.0

Filtered Profit: 0.0

===== Business Insights =====

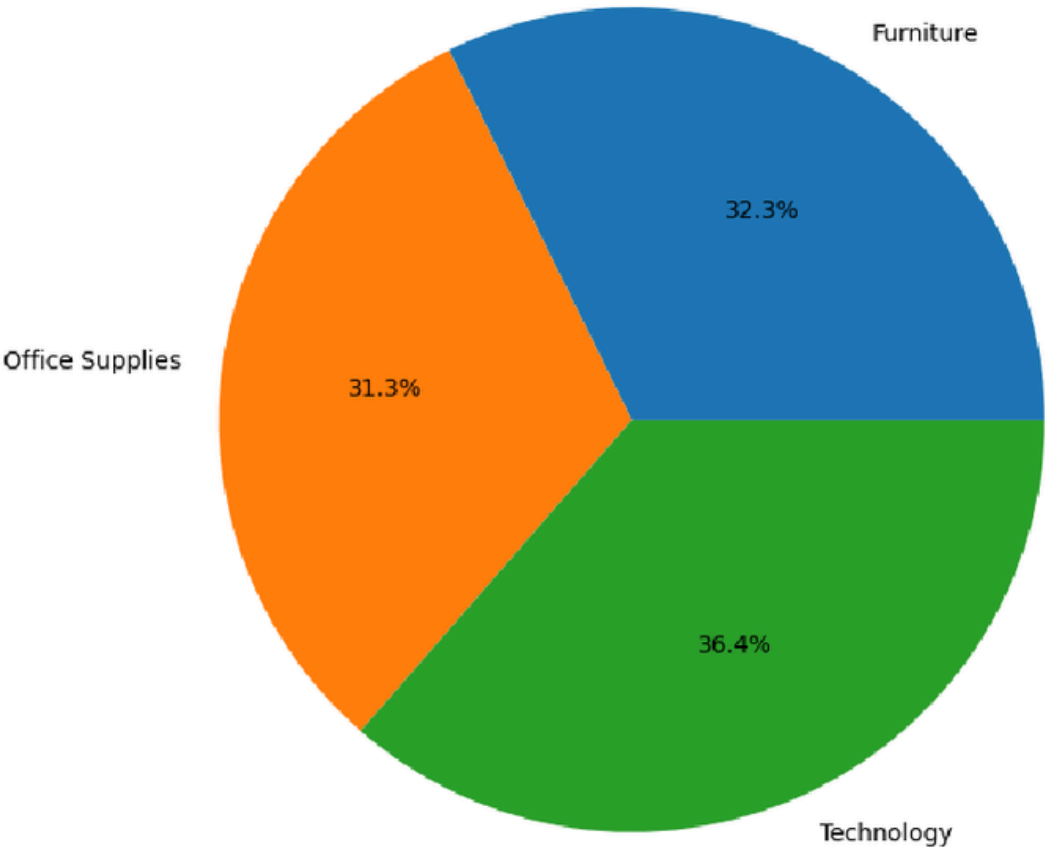
Top Sales Region: West

Highest Region Sales: 725457.82

Top Sales Category: Technology

Highest Category Sales: 836154.03

Sales by Category



Correlation Heatmap

